

1 Introduction

Vocal folds play an integral part in communication by phonation. This lab aims to explore how the vocal folds work to generate sound. More specifically, this lab aims to quantitatively investigate speech production using the Mucosal Wave Model. This lab also aims to explore the role of lung pressure to ensure self-sustained for phonation and try improving the limitations of the original Mucosal Wave Model.

2 Background

2.1 Laryngoscopy

In a medical context, laryngoscopy is the process in which the larynx is examined. However, in context of this lab, this method is used to understand the behaviour of vocal folds and thus motivate a derivation of a mathematical model to explain this behaviour.

In a laryngoscopy, the vocal folds are vibrating very quickly. Videostroboscopy is a common method used to capture the high frequency motion of vocal folds. The tool is able to capture slow motion vibrations of the vocal folds. First, using a microphone, the natural frequency of the voice is measured. Then the images of the vocal folds are sampled at a frequency slightly lower than the natural frequency of the voice. By doing so, it is possible to distinctively capture the phases of the vibration of the vocal fold. The natural frequency of the voice is a baseline at which the folds are vibrating. By sampling at a rate lesser than this frequency, we effectively capture a sparse series of images for the slow-motion vibration.[1][2]

Using videostroboscopy, a very interesting result can be seen. There is a phase offset between the vibrations of the backward vocal chord and forward vocal chord. This helps create the appropriate geometry for the model(as outlined in the next subsection). This phase offset is seen in Figure 1.

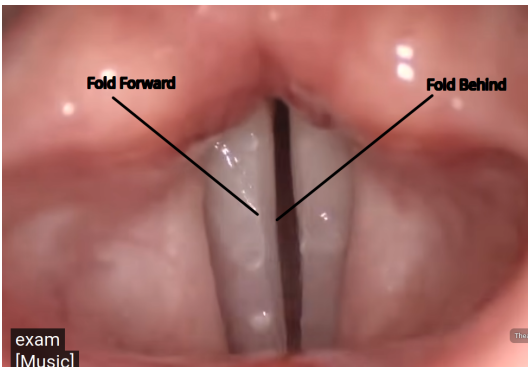


Figure 1: As the figure demonstrates, there is a phase offset between the back fold and the front fold. It is clear that the folds behind leads the folds up in the front.[3]

2.2 Mucosal Wave Model

2.2.1 Model Motivation and Geometry

An initial and more simplistic model for the vocal fold vibration was provided by Flanagan and Landgraf. In this model, they assumed a single degree of freedom and vocal folds open and close in phase in presence of a lung pressure by virtue of the Venturi effect. Although this model does describe some dynamics, it does fail and sometimes predicts non-oscillatory behavior.[4] Furthermore, the model assumes that the folds vibrate in phase, which as shown earlier, is not the case.

The mucosal wave model is thus developed accounting for this phase difference. Instead of one degrees of freedom, the model is developed from two degrees of freedom system. The 2 phase geometry can be shown in figure 2.

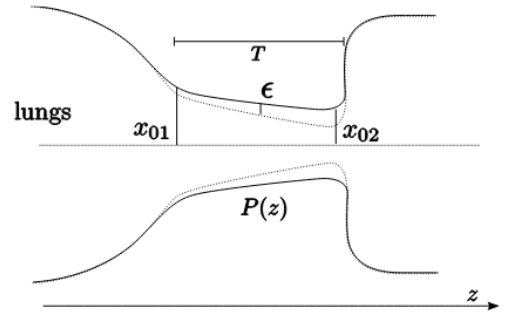


Figure 2: There are 2 folds, the subglottal fold near the lung whose motion under external pressure $P(z)$ is characterized by $x_1(t)$ and the supraglottal fold near the mouth whose motion is characterized by $x_2(t)$. The solid line are the initial conditions of the system i.e. under no external force: $x_1 = x_{01}$ and $x_2 = x_{02}$. The distance between the folds is T and the width is W . [4]

Both $x_1(t)$ and $x_2(t)$ follow the wave equation.[5] Therefore using the D'Alembert form $x_1(t) = x_{01} + f(t + \tau)$ and $x_2(t) = x_{02} + f(t - \tau)$ with $\tau = \frac{T}{2c}$ where c is the speed of the wave. The τ term controls the phase shift between the forward and backward folds.

To aid in the analysis, the system can be decoupled by working with the mean position of the fold and the effective mass, spring constant, and damping of the system. It is assumed the size of the folds vary linearly [6] i.e. at a distance z from x_1 the size of the fold is

$$x_z(z, t) = z \left(\frac{x_2(t) - x_1(t)}{T} \right) + x_1(t)$$

At the mean position $x(z = \frac{T}{2}, t) = \frac{1}{2} (x_1(t) + x_2(t))$. Therefore, the relative displacement of the system is

$$\epsilon(t) = x \left(z = \frac{T}{2}, t \right) - x \left(z = \frac{T}{2}, 0 \right)$$

$$\epsilon(t) = \frac{1}{2} ((x_1(t) - x_{01}) + (x_2(t) - x_{02}))$$

Using this relation and that τ is small, x_1 and x_2 and the respective areas can be expressed as a function of the initial condition, ϵ and $\dot{\epsilon}$ and not on f . For small τ $f(t + \tau) = f(t) + \tau \frac{df}{dt}$

$$\epsilon = f(t), \quad x_1 = x_{01} + f(t) + \tau \frac{df}{dt}, \quad x_2 = x_{02} + f(t) - \tau \frac{df}{dt}$$

Using $\epsilon = f(t)$ and $\dot{\epsilon} = \frac{df}{dt}$

$$x_1 = x_{01} + \epsilon + \tau \dot{\epsilon}, \quad x_2 = x_{02} + \epsilon - \tau \dot{\epsilon}$$

The respective areas are $a_i = 2Wx_i$. Therefore

$$a_1 = 2W(x_{01} + \epsilon + \tau \dot{\epsilon}), \quad a_2 = 2W(x_{02} + \epsilon - \tau \dot{\epsilon})$$

2.2.2 Force Analysis

Using the effective Mass (M) per unit area, Damping (B), Spring Constant (K) per unit area, the dynamics of the system can be expressed as:

$$M\ddot{\epsilon} + B\dot{\epsilon} + K\epsilon = P_g$$

where P_g is the average pressure over the glottis from P_z .

$$P_g = \frac{1}{T} \int_0^T P(z) dz$$

Therefore, the next step is to find an expression for $P(z)$. Figure 3 shows a simplified model for the hydrodynamics.

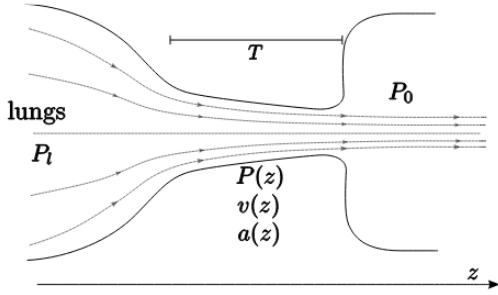


Figure 3: Hydrodynamics within the glottis.[4]

Assuming inlet area to be large, the inlet velocity $v(0)$ is close to 0. Furthermore, it is assumed that Bernoulli is valid within the glottis. Taking the central streamline:

$$P_l = P(z) + \frac{1}{2}\rho v^2(z)$$

It is assumed that the exit is at ambient pressure P_0 , therefore streamlines are straight and parallel with uniform pressure.

$$P_l = P_0 + \frac{1}{2}\rho v_2^2$$

Using these relations:

$$P(z) = P_0 + \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v^2(z)$$

Furthermore, from incompressibility and mass conservation:

$$v(z)a(z) = v_2 a_2 \implies v(z) = v_2 \frac{a_2}{a(z)}$$

Therefore

$$P(z) = P_0 + \frac{1}{2}\rho v_2^2 \left(1 - \frac{a_2^2}{a^2(z)}\right)$$

Since $P_l = P_0 + \frac{1}{2}\rho v^2$

$$P(z) = P_0 + (P_l - P_0) \left(1 - \frac{a_2^2}{a^2(z)}\right)$$

What is quite nice about this form is that it only depends on the geometry (which has been modelled in the previous section), lung pressure, and ambient pressure. There is no need to know the velocities. Furthermore, the average pressure can now be easily calculated using the derived models.

$$a(z) = 2Wx(z) = 2W \left[z \left(\frac{x_2 - x_1}{T} \right) + x_1 \right]$$

Using this $a(z)$ the average pressure can be easily calculated.

$$P_g = \frac{1}{T} \int_0^T P_0 + (P_l - P_0) \left(1 - \frac{a_2^2}{a^2(z)}\right) dz$$

$$P_g = P_0 + \frac{P_l - P_0}{T} \int_0^T 1 - \frac{a_2^2}{4W^2(mz + c)^2} dz$$

where $m = \frac{x_2 - x_1}{T}$ and $c = x_1$

$$P_g = P_0 + \frac{P_l - P_0}{T} \left[z + \frac{a_2^2}{4W^2} \left(\frac{-1}{m(mz + c)} \right) \right]_0^T$$

$$P_g = P_0 + \frac{P_l - P_0}{T} \left[T + \frac{a_2^2}{4W^2} \left(\frac{-1}{m(mT + c)} - \frac{-1}{mc} \right) \right]$$

From the previous expression for $x(z) = mz + c$ $x(T) = x_2$ and $x(0) = x_1$

$$P_g = P_0 + \frac{P_l - P_0}{T} \left[T + \frac{a_2^2}{4W^2} \left(\frac{-1}{mx_1} + \frac{1}{mx_2} \right) \right]$$

$$P_g = P_0 + \frac{P_l - P_0}{T} \left[T + \frac{a_2^2}{4W^2} \left(\frac{x_1 - x_2}{mx_1 x_2} \right) \right]$$

To simplify further $4W^2 mx_1 x_2 = m(2Wx_1)(2Wx_2) = ma_1 a_2$ is used

$$P_g = P_0 + \frac{P_l - P_0}{T} \left[T + a_2 \left(\frac{x_1 - x_2}{ma_1} \right) \right]$$

On expanding $m = \frac{x_2 - x_1}{T}$

$$P_g = P_0 + \frac{P_l - P_0}{T} \left[T - T \frac{a_2}{a_1} \right]$$

Therefore the desired average pressure is obtained:

$$P_g = P_0 + (P_l - P_0) \left(1 - \frac{a_2}{a_1}\right)$$

On gauging it, i.e. $P_0 = 0$

$$P_g = P_l \left(1 - \frac{a_2}{a_1}\right)$$

There are some key assumptions made in this approach of obtaining the average pressure. The most significant one being that Bernoulli Principle is valid throughout the glottis. This has experimentally proven to be invalid with the flow becoming turbulent near the exit. This limitation is addressed in Titze's approach wherein more pressure stations are introduced and the transglottal pressure is given by $k_t P_{k2}$ where $k_t = k_c - k_e$ is the transglottal pressure recovery (k_c is pressure drop coefficient and k_e is a pressure gain within turbulent region) and P_{k2} is the kinetic pressure. In their model, $P_g = P_l \frac{1}{k_t} \left(1 - \frac{a_2}{a_1} - k_e\right)$ thus accounting for the turbulent in

the region.[6] Setting $k_e = 0$ i.e. no turbulent gain and $k_t = 1$ i.e. no kinetic pressure loss or gain, the expression derived above is obtain. For the rest of the handout, the ideal derivation shall be used.

To obtain the equation of motion in ϵ , P_g should be expressed as a function of ϵ . This is quite straightforward since

$$a_1 = 2W(x_{01} + \epsilon + \tau\dot{\epsilon}), \quad a_2 = 2W(x_{02} + \epsilon - \tau\dot{\epsilon})$$

Therefore

$$P_g = P_l \left(\frac{a_1 - a_2}{a_1} \right) = P_l \left(\frac{x_{01} - x_{02} + 2\tau\dot{\epsilon}}{x_{01} + \epsilon + \tau\dot{\epsilon}} \right)$$

Define $\Delta x_0 = x_{01} - x_{02}$, the desired force balance equation is obtained.

$$M\ddot{\epsilon} + B\dot{\epsilon} + K\epsilon = P_l \left(\frac{\Delta x_0 + 2\tau\dot{\epsilon}}{x_{01} + \epsilon + \tau\dot{\epsilon}} \right)$$

This is a non-linear differential equation hence RK(4) shall be leveraged into simulating the analysis.

2.3 Numerical Methods: RK(4)

Runge Kutta methods are a common method of solving complicated system of differential equations with order of error $O(\Delta t^4)$. However, what particularly makes them appealing is their numerical stability.

To work with RK-4, it is imperative to write them in a state space form. [4]

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, t)$$

As an example, consider the system:

$$M\ddot{\epsilon} + B\dot{\epsilon} + K\epsilon = 0$$

The state space model can be expressed as:

$$\frac{d}{dt} \begin{pmatrix} \epsilon \\ \dot{\epsilon} \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \begin{pmatrix} \dot{\epsilon} \\ \frac{1}{M} (-B\dot{\epsilon} - K\epsilon) \end{pmatrix}$$

It is possible to compare this system to the analytical solution and then the performance of RK-4 can be compared. The analytical solution is as follows

$$\epsilon(t) = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$$

$$\lambda_1 = \frac{-B}{2M} + \frac{\sqrt{B^2 - 4MK}}{2M}, \quad \lambda_2 = \frac{-B}{2M} - \frac{\sqrt{B^2 - 4MK}}{2M}$$

The nature of the solution is dependent on the values of the paramaters involved. Specifically on the $B^2 - 4MK$ parameter. $B^2 > 4MK$ is an overdamped system and hence follows the exponential. In practice it is expected to be decaying. $B^2 = 4MK$ is a critically damped system. $B^2 < 4MK$ is an oscillatory system with natural frequency $\sqrt{\frac{K}{M}}$, damping $\frac{B}{2\sqrt{MK}}$ and damped natural frequency: $\sqrt{4MK - B^2}$. To study the effectiveness of RK 4 methods, 2 plots with different parameters were considered with initial condition $\epsilon(t=0) = 10^{-6}$ and $\dot{\epsilon}(t=0) = 0$. The mean squared error between the analytic and numerical solution was calculated. To further validate the results, the analytic plot was plotted and random selection of the RK 4 was plotted. These results are shown in figure 3.

Figure 4a shows the analytical solution for a slightly under-damped system. The mean squared error between the analytical solution and the solution for ϵ predicted by the RK4 method is of the order 10^{-20} and for $\dot{\epsilon}$ in order of 10^{-15} . The sampled points from the RK4 implementation also cleanly

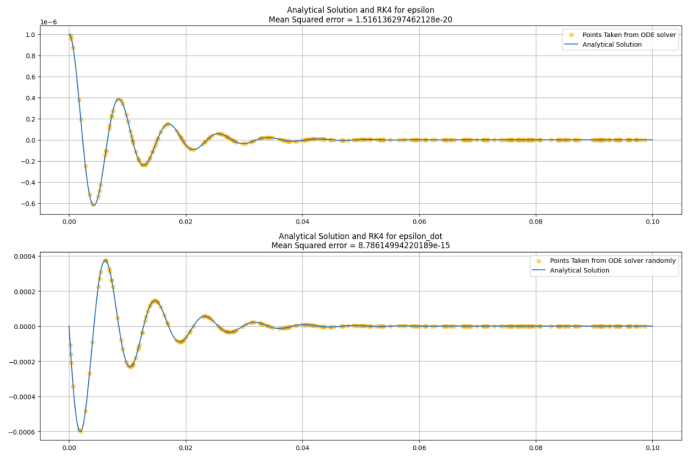


Figure 4(a) - Under Damped System

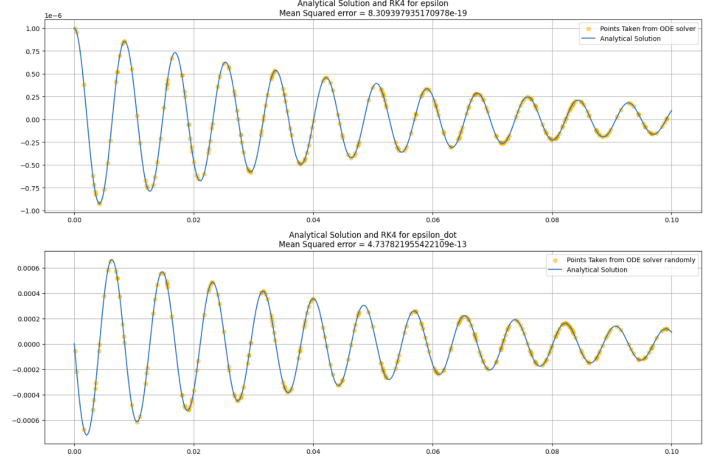


Figure 4(b) - Heavily Under Damped System

Figure 4: Performance of RK4 for different parameter settings

align with the analytical solution. To test the robustness, a heavily under damped system was also considered with mean squared error in ϵ being of order 10^{-19} and $\dot{\epsilon}$ being of order 10^{-13} and the sampled points aligning cleanly with the analytical solution.

The previous results clearly demonstrate that RK(4) is a very powerful numerical method to obtain small errors in computations. Although methods such as RK-4 etc help increase accuracy, they are not symplectic integrators i.e. the analytical solution would drift away from the true solution for long simulations. Methods such as verlet integrators although have a slow convergence are symplectic which means they do not drift away from the true solution. <https://github.com/scipy/scipy/issues/12690> this github issue talks about this as well.[2]

In context to the mucosal wave model, the state space model can be written as:

$$\frac{d}{dt} \begin{pmatrix} \epsilon \\ \dot{\epsilon} \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \begin{pmatrix} \dot{\epsilon} \\ \frac{1}{M} \left[P_l \left(\frac{\Delta x_0 + 2\tau\dot{\epsilon}}{x_{01} + \epsilon + \tau\dot{\epsilon}} \right) - B\dot{\epsilon} - K\epsilon \right] \end{pmatrix}$$

Using this state space representation and RK4, some results are generated computationally.

2.4 Physiological Constants

These are taken from the handout directly and are worked in CGS units.[4]

1. Fold mass per unit Area: $M = 0.45 \text{ g/cm}^2$
2. Tissue Stiffness: $K = 250 \text{ kdyne/cm}^3$
3. Damping (Viscous Loss): $B = 100 \text{ dyne s/cm}^3$

4. Rest Fold Separation: $x_{01} = 0.1$ cm, $x_{02} = 0.09$ cm, $\Delta x = 0.01$ cm
5. Fold Width: $W = 1.4$ cm
6. Fold Length: $T = 0.4$ cm
7. Mucosal Speed: $c_w = 100$ cm/s, $\tau = 0.002$ s

3 Results and Discussion

3.1 Model Implementation

The following is the code used to implement the model.

```
def muc(x, t, p):
    x1, x2 = x[0], x[1] # Store (eps, d(eps)/dt)
    dx1 = x2 #dx/dt = ((eps_dot, eps_ddot)
    # M*eps_ddot + B*eps_dot + K*eps =
    # P_l * (diff_x + 2*tau*eps_dot)/
    # (x01 + eps + tau * eps_dot)

    rhs = p['P1'] * ((p['dx0'] + 2*p['tau']*x2)/
    (p['x01'] + x1 + p['tau']*x2))
    dx2 = 1/p['M'] * (rhs - p['B']*x2 - p['K']*x1)
    return([dx1, dx2])

def run_lab(x0, T, p):
    t = np.linspace(0, T, 1000)
    dxdt = lambda x, t : muc(x, t, p)
    x_t = integrate.odeint(dxdt, x0, t)
    return(t, x_t)

# Create parameters
p = dict(P1=1500, M=0.45, K=250000, B=100,
        x01=0.1, dx0=0.01, tau=0.002)
t, x = run_lab([1e-6, 0.0], 0.1, p)
eps, eps_dot = x[:, 0], x[:, 1]

fig, ax = plt.subplots(2, figsize=(15, 10))

ax[0].plot(t, eps, label='Epsilon')
ax[0].set_title('Epsilon plot in cm')
ax[0].legend()

ax[1].set_title('Epsilon dot plot in cm/s')
ax[1].plot(t, eps_dot, label='Epsilon Dot')
ax[1].legend()
plt.show()
```

Figure 5 showcases the results for the implementation of the wave model. From the plot, it can be seen that under these

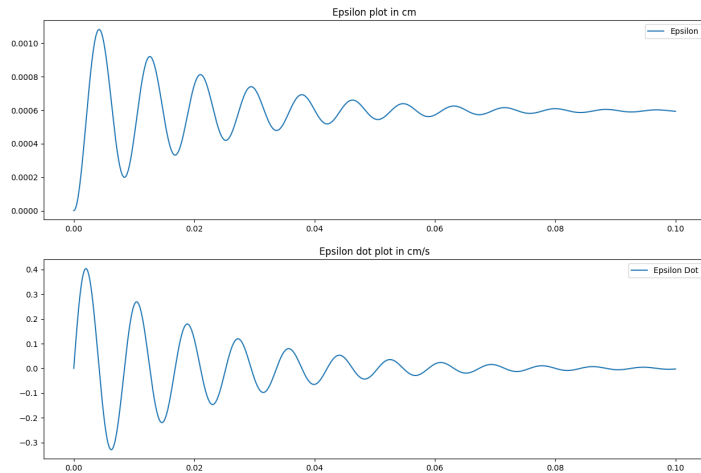


Figure 5: Mucosal Wave Model

conditions, the system is underdamped causing the oscillations to decay and the system to reach a constant stable steady state. In practice, under an influence of a lung pressure the oscillations can be sustained for a prolonged period of time. To help understand resolve, it is important to study how the model behaves on changing the parameters.

3.2 Sensitivity Analysis

The sensitivity analysis was done for all the parameters in the model. The interpretation of the results was done based on ϵ , $\dot{\epsilon}$ and the frequency of the oscillation. To calculate the frequency response a FFT was used. Given this system is non linear, a simple FFT may not capture the harmonic behavior and higher order peak frequencies, and may have some more uncertainties and aliasing artifacts; however, it should suffice for a rudimentary analysis for the frequency especially since the plot seems periodic. In general it is possible to numerically estimate the complete General Frequency Response Function using Volterra series/kernels but this remains outside the scope of this lab. Therefore for the sake of simplicity, only a FFT was included to study the single peak frequency that governs the periodic nature of the system.[2]. The code for the sensitivity analysis and interactive plots can be found in the github repo in the appendix.

After running and investigating the sensitivity analysis, the following results were obtained [2]:

1. Mass : From the plots, mass is inversely related to frequency. As the mass rises, the peak frequency decreases.
2. Damping : It controls the duration of the vibration and has no relation to the position of the maximum frequency. It does reduces the magnitude of the frequency plot which is as expected. Furthermore, the damping also seems to control the amplitude of the oscillations and its stability. As damping increases, the amplitude of the time-series data decreases. For really small values of damping, the system becomes unstable. Despite the system being non-linear, the damping does seem to control the half power bandwidth of the system as well with smaller damping making it narrower. However, the relation is not as simple as the linear case $\zeta \approx \frac{\Delta\omega}{\omega_n}$. A general frequency response may help yield a much more accurate behaviour.
3. Spring Constant : It controls the frequency of the produced sound. Increasing K increases the peak frequency at the cost of decreasing the magnitude of the peak frequency and amplitude of the time-series data.
4. Lung Pressure : A higher lung pressure reduces the effect of damping and increases the gain at the maximum frequency. It also increases the amplitude of the vibrations. For smaller lung pressures the solution becomes a simple SHM problem, which is particularly interesting. This suggests, that the lung pressure plays a critical role in the steady state behavior of the system. Small relative changes in lung pressure cause significant changes in the steady state value of the system. Furthermore, there must be another critical lung-pressure apart from 0 where the system is marginally stable (the system transitions from a stable system to an unstable one suggesting a critical pressure).
5. x_{01} : Seems to control damping but is much more sensitive as opposed to lung pressure. Smaller x_{01} reduces damping but makes the system more unstable.
6. Δx_0 : Seems to control the final steady state value of the system (if its not unstable)
7. τ : Seems to control the damping of the system with higher tau decreasing the damping.

To summarize, the frequency is controlled by the mass and the spring constant while the stability and damping is controlled by the remaining parameters.

The behavior of the lung pressure is interesting. Consider figure 6, where the lung pressure is large and the system damping is small. The system becomes unstable and the oscillations are sustained with increasing amplitudes. Compare this to figure 5 where the lung pressure is small and the system is damped, preventing any oscillations.

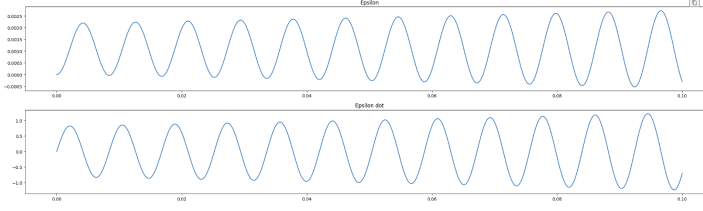


Figure 6: Mucosal Wave Model: Large lung pressure

Using these ideas, it is possible to understand the general mechanics for vocal production. Without sufficient external lung pressure, the amplitude and duration of the oscillations are small (of the order of milliseconds) due to the damping present within the vocal systems. However, on adding external lung pressure it is possible to increase the duration and amplitude of the oscillations. Eventually if the lung pressure is large enough it completely counters the damping of the system and the oscillations can sustain themselves and help sustain phonation. In other words, this external lung pressure adds energy to the system to facilitate sustained oscillations and produce voice for a longer period of time. This is precisely why the system becomes unstable for smaller damping: the lung pressure is now large enough to facilitate self-sustaining oscillations. This interpretation seems in line with literature.[2][6]

3.3 Critical Pressure

3.3.1 Analytic Derivation

As mentioned before, there must be a critical pressure at which the system is marginally stable, i.e. purely oscillatory. This occurs when the lung pressure is equal to the overall damping. The overall damping depends on the damping factor of the folds and the geometry of the system with some small influence by the spring constant. Therefore critical pressure $P_c = f(K, B, \tau, x_{01}, \Delta x_0)$. However, the influence of K on the amplitude of oscillations was small therefore $P_c \approx f(B, \tau, x_{01}, \Delta x_0)$

Recall the Mucosal Wave Model:

$$M\ddot{\epsilon} + B\dot{\epsilon} + K\epsilon = P_l \left(\frac{\Delta x_0 + 2\tau\dot{\epsilon}}{x_{01} + \epsilon + \tau\dot{\epsilon}} \right)$$

Let $\epsilon(t) = \bar{\epsilon} + \delta\epsilon(t)$ i.e. $\epsilon(t)$ oscillates around a mean value. To find the mean value $\bar{\epsilon}$, set $\delta\epsilon(t) = 0 \implies \ddot{\epsilon} = \dot{\epsilon} = 0$

$$\bar{\epsilon} = \frac{-Kx_{01} + \sqrt{K^2x_{01}^2 + 4P_lK\Delta x_0}}{2K}$$

Consider $\delta\epsilon(t)$ to be small. The model can now be written as

$$M\delta\ddot{\epsilon} + B\delta\dot{\epsilon} + K(\bar{\epsilon} + \delta\epsilon) = P_l \left(\frac{\Delta x_0 + 2\tau\delta\dot{\epsilon}}{x_{01} + \bar{\epsilon} + (\delta\epsilon + \tau\delta\dot{\epsilon})} \right)$$

The RHS can be linearized by taking $\delta\epsilon + \tau\delta\dot{\epsilon}$ as small

$$M\delta\ddot{\epsilon} + B\delta\dot{\epsilon} + K(\bar{\epsilon} + \delta\epsilon) = \frac{P_l(\Delta x_0 + 2\tau\delta\dot{\epsilon})}{x_{01} + \bar{\epsilon}} \left(1 - \frac{\delta\epsilon + \tau\delta\dot{\epsilon}}{x_{01} + \bar{\epsilon}} \right)$$

On further simplification and neglecting second order term (since they are small) obtain

$$M\delta\ddot{\epsilon} + B'\delta\dot{\epsilon} + K'\delta\epsilon = P'$$

$$B' = B - P_l\tau \left(\frac{2x_{01} + 2\bar{\epsilon} - \Delta x_0}{(x_{01} + \bar{\epsilon})^2} \right)$$

$$K' = K + \frac{P_l\Delta x_0}{(x_{01} + \bar{\epsilon})^2}$$

$$P' = \frac{P_l}{x_{01} + \bar{\epsilon}} - K\bar{\epsilon}$$

Given at the critical pressure P_c the system is purely oscillatory with no energy decay, the effective damping $B' = 0$. Therefore on simplifying for $\bar{\epsilon}$

$$\bar{\epsilon} = \frac{P_c\tau - Bx_{01} + \sqrt{P_c^2\tau^2 - BP_c\tau\Delta x_0}}{B}$$

There are 2 equations for $\bar{\epsilon}$: one obtained from steady state analysis and one for when the system is at critical pressure. On equating both, the critical pressure can be found. However, this is slightly challenging to do for a general case. To simplify the calculation, consider the physiological property that K is significantly large (see physiological constants in background). Therefore from steady state: $\bar{\epsilon} \sim \frac{-Kx_{01} + \sqrt{K^2x_{01}^2}}{2K} = 0$ Therefore since $\bar{\epsilon}$ is 0, the critical pressure is obtained:

$$P_c \approx \frac{Bx_{01}^2}{\tau(2x_{01} - \Delta x_0)}$$

This is in line with what was expected as described in the previous section, i.e. $P_c = f(B, x_{01}, \Delta x_0, \tau)$ and in line with Titze work (apart from the missing turbulent gain term).[6] The complete solution without any approximation and directly equating the 2 $\bar{\epsilon}$ is very challenging to solve algebraically as it forms a quartic. The next section will demonstrate that the derived estimate for P_c is close to the numerical estimate.

Using the physiological constants, the theoretical critical pressure is calculated to be 2631.58 dyne/cm².

3.3.2 Numerical Estimation

The following code was used to calculate the critical pressure.

```
p = dict(Pl=1500, M=0.45, K=250000, B=100,
x01=0.1, dx0=0.01, tau=0.002)

def obtain_diff(p):
    t,x = run_lab([1e-6,0],T=0.1,p=p)
    eps = x[:,0]
    eps_peaks_indices = find_peaks(eps)[0]

    first_index = eps_peaks_indices[0]
    second_index = eps_peaks_indices[1]
    first,second = eps[first_index],eps[second_index]

    diff = first - second
    return diff

def estimate_critical_pressure(p,target=0):
    Pl_start, Pl_end = 1500, 3000
    diff = float('-inf')
    max_iter = 1000
```



```

iter_count = 0
while np.abs(diff-target)>=1e-8
    and iter_count <max_iter:
        Pl_mid = (Pl_start + Pl_end)/2
        p['Pl'] = Pl_mid

        diff = obtain_diff(p)
        if diff > 0:
            Pl_start = Pl_mid
        elif diff < 0:
            Pl_end = Pl_mid
        else:
            print(f'Converged with no error')
            return Pl_mid
        iter_count+=1

return Pl_mid, diff

critical_pressure, critical_diff =
estimate_critical_pressure(p)
print(f'Critical Pressure: {critical_pressure}')

p['Pl'] = critical_pressure
t, x = run_lab([1e-6,0.0],0.1,p)
eps, eps_dot = x[:,0], x[:,1]

fig, ax = plt.subplots(2, figsize=(15,10))

ax[0].plot(t, eps, label='Epsilon')
ax[0].set_title('Epsilon plot in cm')
ax[0].legend()

ax[1].set_title('Epsilon dot plot in cm/s')
ax[1].plot(t, eps_dot, label = 'Epsilon Dot')
ax[1].legend()
plt.show()

```

The method in this approach was that if $P_l < P_c$ the difference in the amplitudes of the 1st and 2nd maxima is positive (first peak amplitude - second peak amplitude). If $P_l = P_c$ the difference is 0 and if $P_l > P_c$ is negative. Keeping this logic in mind, a binary search was used to numerically estimate the critical pressure. Figure 7 showcases the response when the lung pressure is at the critical pressure.

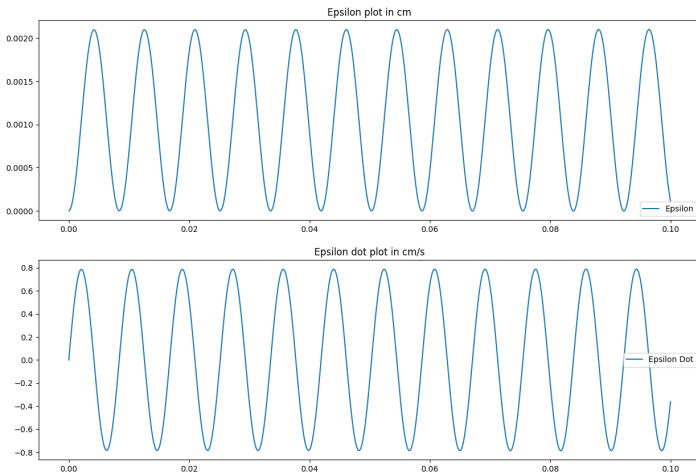


Figure 7: Oscillatory behavior

The algorithm described above converged after 13 iterations with a numerically estimated critical pressure of 2658 dyne/cm². At this critical pressure, the oscillations are self

sustained and the system is marginally stable. To approximate the precision of the results, consider a small percent deviation in pressure, and measure the percent change in difference in successive amplitudes. Then from that estimate how much pressure change is needed for a 10% deviation in the difference between successive peaks. To achieve 10% change in deviation of the 2 successive amplitudes, the percent change in pressure needed is in the order of 10^{-5} suggesting the result to be highly precise.

The theoretical pressure is 2631 dyne/cm² and thus the relative percent error between them is 1.03%. This error is not too significant and is a result of taking $\bar{\epsilon} = 0$.

It is thus reasonable to state the numerical approach aligns with the analytical derivation.

3.4 Amplitude Growth for Large Lung Pressures

3.4.1 Large Lung Pressure without Feedback

For large lung pressures, the system becomes highly unstable: the amplitudes keep growing. This does not make much practical sense since it implies that the energy produced in phonation keeps increasing indefinitely or until the glottis is forcefully closed. It also suggests that the folds do not interact with each other and there are no physical constraints stopping the system from behaving in this manner. From a numerical simulation, it does mean that for large lung pressures, the RK4 implementation breaks. This behavior can be seen in figure 8.

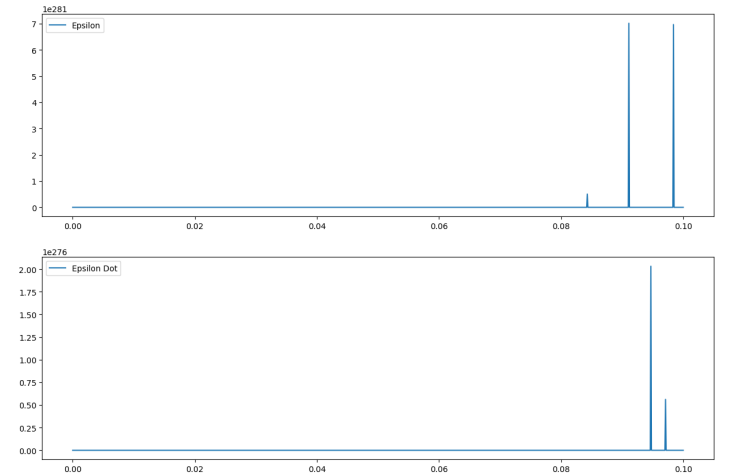


Figure 8: Response for large lung pressure (3500 dyne/cm²)

As the plot above shows, the integrator eventually collapses for large pressures. Given the amplitudes of the peaks are upto large orders, it must be due to division by 0 during the integrator process. This is further evidenced from the fact that the system becomes unstable as the amplitude of the oscillations keep rising.

The behavior in the actual phonation system is considerably different. When ϵ gets larger than the fold separation Δx_0 there is a much larger dissipation in energy thus the damping becomes significant.[4] This thus invites the idea of a non linear control system with feedback to help ensure that ϵ does not grow indefinitely.

3.4.2 Saturate Amplitude Growth

The objective is to saturate the amplitude and stop indefinite growth. As mentioned before, the damping is sensitive to ϵ

and hence the original system can be modified to:[4]

$$M\ddot{\epsilon} + Bg(\epsilon)\dot{\epsilon} + K\epsilon = P_l \left(\frac{\Delta x_0 + 2\tau\dot{\epsilon}}{x_{01} + \epsilon + \tau\dot{\epsilon}} \right)$$

Using the function g it is possible to modify the damping accordingly. It is expected that $g(\epsilon)$ increases when ϵ is larger than the Δx_0 as the folds touch there is a larger impact of damping. Therefore, a rough form for $g(\epsilon)$ is

$$g(\epsilon) = \begin{cases} 1 & |\epsilon| \leq \Delta x_0 \\ 1 + C(||\delta\epsilon| - \Delta x_0||)^p & \text{otherwise} \end{cases}$$

where $||\dots||_p$ is the p th distance measure (For example $p = 1 \implies \text{distance} = ||\epsilon| - \Delta x_0|$ etc) and $\delta\epsilon$ is the oscillatory term in complex form. Therefore $|\delta\epsilon|$ gives the amplitude of vibrations. The key idea of this proposed function is that when the folds are separated from each other, the system damping is purely given by B . However, when the folds touch, there is an additional damping given by $C(||\delta\epsilon| - \Delta x_0||)^p$. It is possible to write $|\delta\epsilon| = |\epsilon - \bar{\epsilon}|$ where $\bar{\epsilon}$ is the mean displacement.

Similar work was done in the past by Rodrigo Laje, Tim Gardner, and G. B. Mindlin. In their approach they studied relaxation oscillations to model changes in open and close states of the glottis. The open and closed state are related to the amplitude of the vibration. The key ideas is that there are 2 alternating processes occurring at a fast and slow time scale. The slow time-scale is the period in which the system approaches an equilibrium and the fast time scale corresponds to shift in equilibrium.[7] The phase space equations for relaxation oscillations are $u' = v - u^3 + u$ and $v' = -\epsilon u$ where u is the fast variable and v is the slow variable. Figure 9 is a phase portrait of the system when ϵ is small. When ϵ is small, the phase portrait is similar to a bifurcation diagram and treating v is the parameter.[8] The bifurcation diagram and phase portrait diagram help understand the system properties, more specifically the stability of the different equilibrium points.[9]

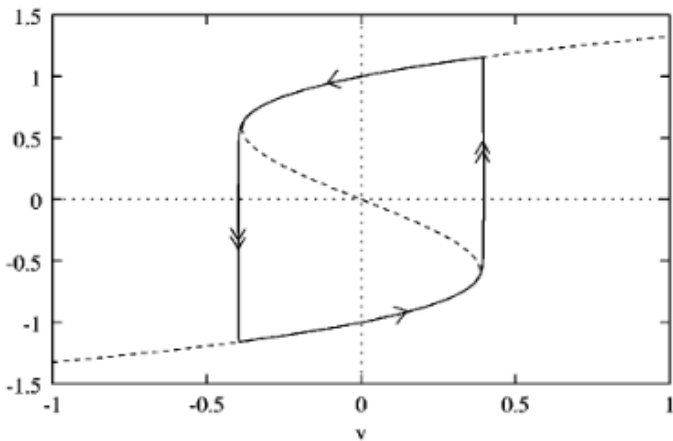


Figure 9: Relaxation Oscillator Phase Portrait

From the figure, it is possible to note that this relaxation oscillatory behavior can explain jumps between 2 states. (see the vertical tangential lines). Therefore, to combine this approach to the glottis, a similar model is built from a Newtonian Context. Let

$$x = u$$

and thus using the phase equations and Newtonian form

$$\ddot{x} = f(\dot{x}, x, t)$$

the dynamical system can be written as

$$\ddot{x} = -\epsilon x + \dot{x}(-3x^2 + 1)$$

From this, the key insight is the $-3x^2\dot{x}$ term which forms the amplitude saturation (non linear dissipative force) while the $-\epsilon x$ term is the "spring force" and $\dot{x}$ is the natural system damping. Using this relation, the authors argued that when $\bar{\epsilon}$ is not negligible, the damping must be increased if ϵ overshoots the mean value. Therefore, $g(\epsilon) = (1 + C(\epsilon - \bar{\epsilon})^2)$ and the final equation is[8]

$$M\ddot{\epsilon} + B(1 + C(\epsilon - \bar{\epsilon})^2)\dot{\epsilon} + K\epsilon = P_l \left(\frac{\Delta x_0 + 2\tau\dot{\epsilon}}{x_{01} + \epsilon + \tau\dot{\epsilon}} \right)$$

Both expressions for $g(\epsilon)$ have the core idea: penalizing overshoot in amplitude. From a numerical solver perspective, it is a bit easier to work with the derivation by Laje et al since there are no piece-wise functions and absolute values to worry about in the implementation. The second order nature allows smoothness and easier calculations. Figure 10 is the implementation of the modified mucosal wave model.

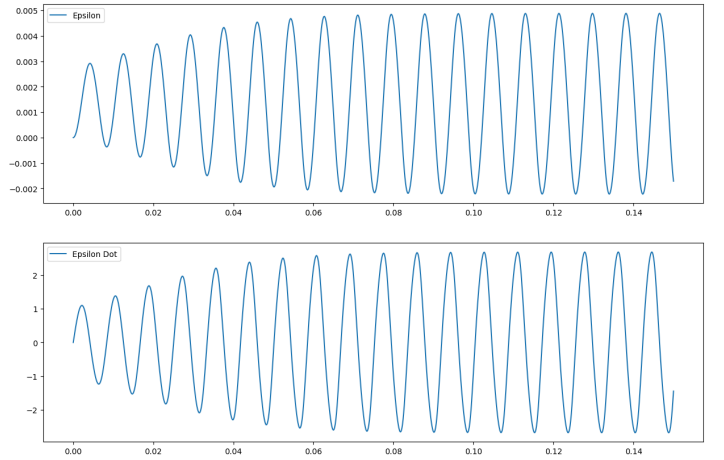


Figure 10: Extreme Lung Pressure (3500 dyne/cm²) with $C = 100000$

Comparing figure 10 to figure 8, it is clear that the amplitude starts saturating and the oscillations are self-sustained at around 0.005cm. Although the choice of C being this large seems counter-intuitive, it does make sense. The squared difference between ϵ and $\bar{\epsilon}$ is of the order 10^{-6} therefore C should be around the order of magnitude to meaningfully alter B for large oscillations. If the value of C is too small, its contribution to the damping is non existent and if it is too large the system behaves like an over damped system.

4 Conclusion

This lab has thus successfully explored how vocal folds worked. To produce self sustained oscillations, a critical lung pressure must be reached and then to ensure that these oscillations form a stable system a non linear damping factor is required as well.

Although this lab was able to explain a working model, the ideal next step would be to verify it using experimental data. Using videostroboscopy as mentioned earlier, it could be possible to take live measurements of ϵ and the model can then be fit.

In context to the actual model that was derived at the end (including the non linear damping), it would be really interesting to see how sensitive the non linear damping gain parameter C is, especially in context to how quickly it causes

the amplitude to saturate. It would be really interesting to study the behaviour of this parameter from a control theory perspective. Furthermore, it would be ideal to use numerical or analytical methods to estimate this parameter as well.

5 Appendix

The following is the code for the lab: https://github.com/Arnav910/3G2_lab

6 References

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